UNIVERSITY OF DERBY
College of Science and Engineering

Practice Paper B

OPERATIONAL RESEARCH
5CM522

DATE: REVISION MATERIAL
TIME ALLOWED: 4 HOURS

Instructions to Candidates

You are required to answer a total of 4 questions.

However, you may attempt all 6 and your best 4 will be considered.

This is a practice paper designed to support preparation for the timed in-class test.

You may use Excel or Python, but your workings, files and interpretations must be clear and well labelled.

You may access code, notes and module files while practising.

You must not use AI tools if you want the practice to reflect the conditions of the real in-class test.

1.

Silverforge Engineering is studying two scheduling problems.

Part A: Flow Shop Scheduling

All five jobs must be processed first on Machine 1 and then on Machine 2.

Job	Machine 1 Time	Machine 2 Time
J_1	7	5
J_2	3	8
J_3	6	4
J_4	2	9
J_5	5	6

Part B: Job Shop Scheduling

Each job has its own machine order and must follow that order exactly.

Job	Sequence on Machines	Processing Times
A	M_1, M_3, M_2	$4, 5, 3$
B	M_2, M_1, M_3	$3, 4, 6$
C	M_3, M_2, M_1	$5, 2, 4$
D	M_1, M_2, M_3	$2, 6, 3$

(a) Use Johnson's Rule to determine an optimal production sequence for the flow shop problem in Part A. Show the main steps. (6 Marks)

(b) Draw a Gantt chart for your sequence and calculate the makespan. (5 Marks)

(c) Compare your Johnson schedule with the original order J_1, J_2, J_3, J_4, J_5 and comment on the improvement. (4 Marks)

(d) Solve the job shop scheduling problem in Part B and state an optimal schedule and the corresponding makespan. You may present the result as a table of operation start and finish times or as a Gantt chart. (6 Marks)

(e) Suppose job C must finish by time 15. Explain how the model changes and discuss the possible effect on feasibility and makespan. (4 Marks)

Total 25 Marks

Continued on Next Page

2.

County Response wants to choose a limited number of patrol bases. The aim is not to cover every district, but to maximise the total population covered while opening at most three bases.

Base	Cost	Population Covered	Districts Covered
A	14	18,000	$\{1, 2, 4\}$
B	11	15,000	$\{1, 3\}$
C	13	21,000	$\{2, 3, 5\}$
D	10	12,000	$\{4, 5\}$
E	12	16,000	$\{2, 5, 6\}$
F	9	14,000	$\{3, 4, 6\}$

The population living in each district is given below.

District	1	2	3	4	5	6
Population	9,000	12,000	14,000	10,000	11,000	8,000

The budget is £31 and at most 3 bases may be opened.

(a) Formulate a binary integer programming model to maximise covered population subject to the budget and base limit. Clearly define any coverage variables you use. (9 Marks)

(b) Solve the model and identify which bases should be opened and the resulting total covered population. (6 Marks)

(c) Explain whether some districts may remain uncovered in an optimal solution and why this is acceptable in this model but not in a set covering model. (4 Marks)

(d) Modify the model so that district 6 must be covered and bases B and F cannot both be chosen. State the revised model changes clearly. (6 Marks)

Total 25 Marks

3.

A university library help desk is modelled as a queueing system. Students arrive randomly and service times are assumed to follow an exponential distribution. You should adapt the SimPy structures used in class, but tailor the model to this setting.

Use the following assumptions:

Quantity	Value
Simulation length	360 minutes
Average service time	7 minutes
Baseline arrival rate	10 students per hour
Peak arrival rate	15 students per hour
Service target	97% wait less than 8 minutes
Priority users	10% of arrivals
Suggested random seed	24

(a) Describe the main elements of a discrete-event simulation model for this system: entities, resources, events, stochastic inputs and performance measures. (5 Marks)

(b) Adapt a SimPy model to represent the baseline system with 2 advisers and an arrival rate of 10 students per hour. Report the total served, average wait, maximum wait and percentage waiting less than 8 minutes. (6 Marks)

(c) Investigate the peak-period scenario with 2, 3 and 4 advisers. Present your results in a clearly labelled table and determine the minimum number of advisers needed to meet the service target. (4 Marks)

(d) Extend the model so that 10% of users are priority users who should be served ahead of standard users. Explain how the model should be changed and discuss the likely effect on standard-user waits. (6 Marks)

(e) Explain why a single run is not enough to support a staffing decision and describe how you would use replications and/or a warm-up period to improve the analysis. (4 Marks)

Total 25 Marks

Continued on Next Page

4.

The operations analyst at Broadleaf Homeware has collected the following quarterly demand data.

Period	Year	Quarter	Demand
1	2019	Q1	420
2	2019	Q2	500
3	2019	Q3	470
4	2019	Q4	560
5	2020	Q1	450
6	2020	Q2	536
7	2020	Q3	500
8	2020	Q4	600
9	2021	Q1	485
10	2021	Q2	575
11	2021	Q3	535
12	2021	Q4	645
13	2022	Q1	522
14	2022	Q2	618
15	2022	Q3	572
16	2022	Q4	694

(a) Using periods 1–12 only, calculate suitable centred 4-quarter moving averages and estimate the trend values where possible. Plot the original data together with the estimated trend. (8 Marks)

(b) Using periods 1–12 only, calculate seasonal indices using both additive and multiplicative decomposition models. (7 Marks)

(c) Use the 2022 data (periods 13–16) as a validation set. Generate forecasts for these four periods using both models and compare their accuracy using one suitable measure such as MAD, MSE or MAPE. (6 Marks)

(d) Using the better model, produce forecasts for the four quarters of 2023 and justify your final model choice. (4 Marks)

Total 25 Marks

5.

Harbour Plastics makes two products each week: casings and inserts.

Each casing requires 4 moulding hours and 1 assembly hour. Each insert requires 2 moulding hours and 3 assembly hours. At most 100 moulding hours and 90 assembly hours are available. No more than 24 inserts can be sold each week.

The weekly contribution is £40 per casing and £34 per insert.

(a) Formulate a linear programming model for this problem. (6 Marks)

(b) Solve the model to determine the production plan that maximises weekly contribution. (6 Marks)

(c) State the binding constraints and the unused capacity in each department. (4 Marks)

(d) A distributor now requires at least 25% of total output to be inserts. Reformulate the model and determine whether the original solution remains feasible. (5 Marks)

(e) If up to 12 extra assembly hours can be purchased at £3 per hour for one week only, determine whether they should be bought. (4 Marks)

Total 25 Marks

Continued on Next Page

6.

Northline Distribution must ship goods from three warehouses to four stores at minimum cost.

Warehouse	S_1	S_2	S_3	S_4
W_1	6	8	10	7
W_2	9	5	7	6
W_3	4	7	6	8

Warehouse	Supply
W_1	80
W_2	100
W_3	90

Store	Demand
S_1	70
S_2	60
S_3	80
S_4	60

(a) Formulate a transportation model for this problem, clearly defining the decision variables, objective function and constraints. (7 Marks)

(b) Solve the model and determine the optimal shipping plan and total transportation cost. Report only the non-zero shipments. (7 Marks)

(c) Resolve the model if warehouse W_2 cannot supply store S_2, and at least 20 units must be shipped from W_1 to S_4. Comment on the impact on total cost. (6 Marks)

(d) Explain one realistic extension that would make this transportation model more useful in practice. (5 Marks)

Total 25 Marks

End of Paper